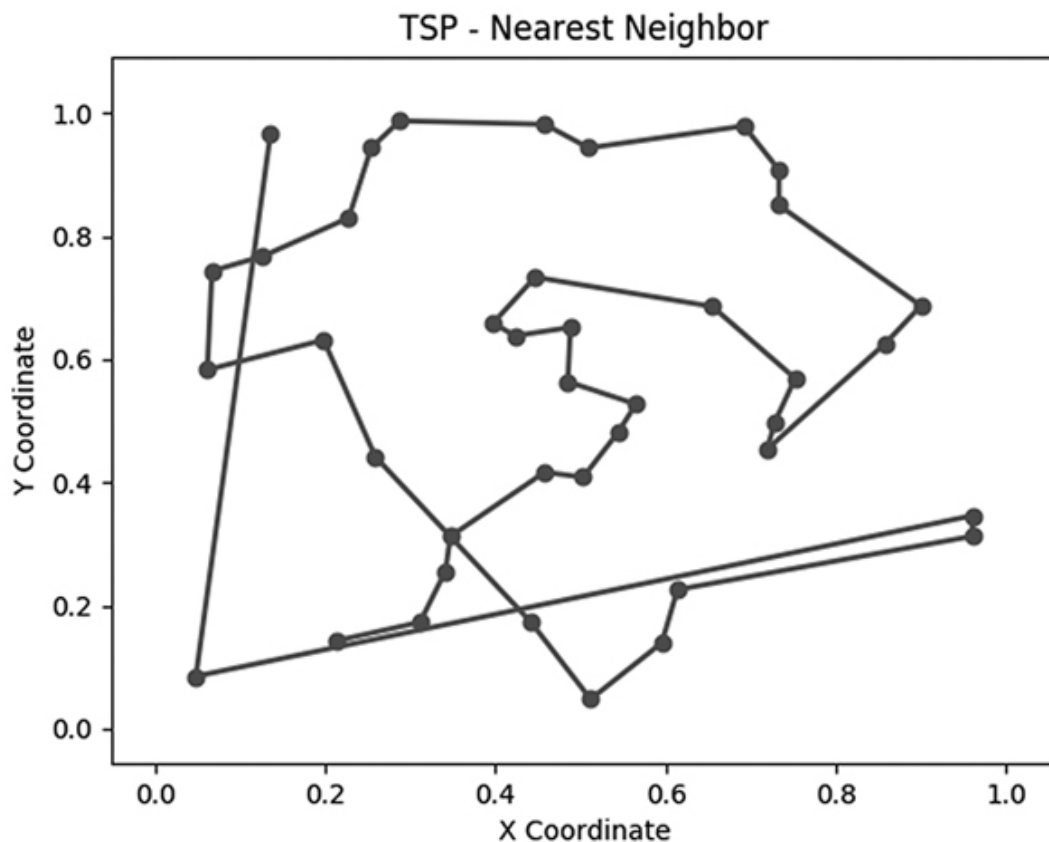




[Figure 6-3](#): The itinerary generated by the nearest neighbor algorithm

We can also check how far the salesman had to travel using this new itinerary:

---

```
print(howfar(genlines(cities,donn(cities,N))))
```

---

In this case, we find that whereas the salesman travels a distance of 16.81 following the random path, our algorithm has pushed down the distance to 6.29. Remember that we're not using units, so we could interpret this as 6.29 miles (or kilometers or parsecs). The important thing is that it's less than the 16.81 miles or kilometers or parsecs we found from the random itinerary. This is a significant improvement, all from a very simple, intuitive algorithm. In [Figure 6-3](#), the performance improvement is evident; there are fewer journeys to opposite ends of the map and more short trips between cities that are close to each other.